

HOW TO MAKE A LEMON BATTERY

Ask an adult to help you get what you need. The screws must be galvanized, which means coated with zinc. LEDs and copper wire with clips can be found in craft or electronics stores. While this experiment is safe, do remember that electricity can be dangerous. And discard the lemons when you have finished the experiment—don't use them for food.



Time
15 minutes



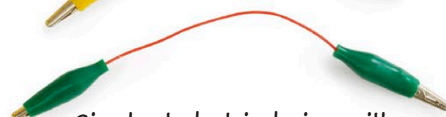
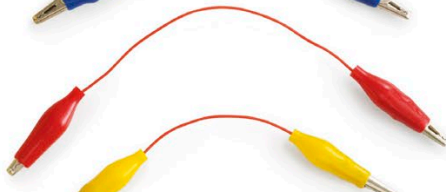
Difficulty
Medium

Warning
Get an adult's help when using the sharp knife

WHAT YOU NEED



Five lemons



Six short electrical wires with clips at each end



Five copper coins



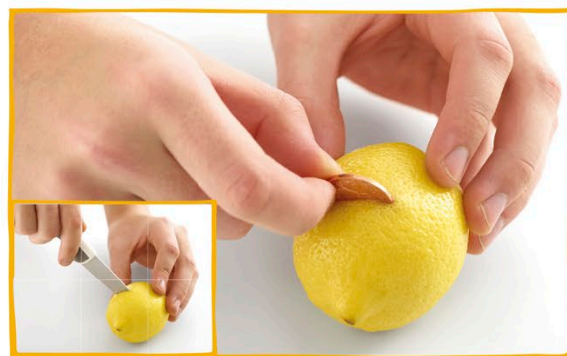
Sharp knife



Five galvanized screws



LEDs (at least one color)



1 With an adult's help, use the knife to make a cut in a lemon, about $\frac{1}{2}$ in (1cm) from the center, and roughly $\frac{3}{4}$ in (2cm) deep. Now push a coin firmly into the slit you have created. Do the same with the other four lemons.



2 About $\frac{1}{2}$ in (1cm) from the center of the first lemon—on the other side to the coin—insert a galvanized screw. Twist it in, clockwise, to secure it in the lemon's flesh. Now repeat with the other four lemons, then arrange the lemons in a circle.